6 A boy with a ball was in a stationary lift. When the lift starts to accelerate upwards at 1.2 m s^{-2}, the boy released the ball from a height of 1.5 m above the floor of the lift.

What is the time taken by the ball to hit the floor of the lift?

A 0.27 s **B** 0.52 s **C** 0.55 s **D** 0.59 s

Ans: B

The relative acceleration of the ball to the lift $= 1.2 + 9.81 = 11.01$ m s^{-2}

Using $\frac{\text{Eqn1}}{\text{Eqn2}}: \frac{s_{y1}}{s_{y2}} = \frac{t_1^{\,2}}{t_2^{\,2}}$, we get

$$1.5 = \frac{1}{2}(11.01)t^2$$

$$t = 0.52 \text{ s}$$